

Stony Brook Institute at Anhui University
2022 Fall Semester
***Fundamentals of Programming* Final Exam (Paper B)**
Reference Answers and Scoring Criteria

I. Multiple Choices (10 items, 1 point each)

No.	1	2	3	4	5	6	7	8	9	10
Answer	A	D	B	B	A	C	A	D	C	C

II. Blank Fillings (10 items, 2 points each)

- (1) `i < nums.length-1`
- (2) `nums[i+1]`
- (3) `mousePressed`
- (4) `rect(mouseX, mouseY, 20, 20)`
- (5) `ellipse(x, y, diam, diam)` or `circle(x, y, diam)`
- (6) `drawCircle(x, y, diam)`
- (7) `P3D`
- (8) `rotateX(PI/4)`
- (9) `loadImage`
- (10) `x + y*width`

III. Corrections (5 items, 2 points each)

- 1. `for (int y=0; y<height; y+=10)`
- 2. `int len = message.length();`
- 3. `Car(int _x, int _y, int _c)`
- 4. `for (int i = 0; i < cars.length; i++)`
- 5. `for (int i = 0; i < pixels.length; i++)`

IV. Programming (5 items, 60 points in total)

- 1. (10 points)
`size(200, 200);`
`String[] stuff = loadStrings("data.txt");`
`int[] data = int(split(stuff[0], ','));`
`background(255);`

```

stroke(255);
int w = width/data.length;
for (int i = 0; i < data.length; i++) {
    fill(data[i]);
    rect(i*w, height-data[i], w, data[i]);
    fill(0);
    text(data[i], i*w, height-data[i]);
}

```

2. (10 points)

```

size(200, 200);
PImage img = loadImage("sunflower.jpg");
loadPixels();
for (int x = 1; x < width; x++) {
    for (int y = 0; y < height; y++) {
        int loc = x + y*img.width;
        color pix = img.pixels[loc];
        int leftLoc = (x - 1) + y*img.width;
        color leftPix = img.pixels[leftLoc];
        float diff = abs(brightness(pix) - brightness(leftPix));
        pixels[loc] = color(diff);
    }
}
updatePixels();

```

3. (10 points)

```

import processing.video.*;
Capture video;

void setup() {
    size(640, 480);
    video = new Capture(this, 640, 480);
    video.start();
}

void captureEvent(Capture video) {
    video.read();
}

```

```
void draw() {
    image(video, 0, 0);
}
```

4. (15 points)

```
String[] headlines = {"I like Processing.",
                      "I like Java, too.",
                      "Do you like Processing and Java?"};
```

```
PFont f;
float x;
int index = 0;
```

```
void setup() {
    size(640, 360);
    f = createFont( "Arial", 16);
    x = width;
}
```

```
void draw() {
    background(255);
    fill(0);
    textFont(f, 16);
    textAlign(LEFT);
    text(headlines[index], x, height-20);
    x = x - 3;
    float w = textWidth(headlines[index]);
    if (x < -w) {
        x = width;
        index = (index + 1) % headlines.length;
    }
}
```

5. (15 points)

```
Ball[] balls = new Ball[100];
```

```
void setup() {
    size(640, 360);
```

```

for (int i = 0; i < balls.length; i++) {
    int x = (int)random(width);
    int y = (int)random(height);
    int size = (int)random(16, 32);
    color c = color(random(255), random(255), random(255));
    int speed = (int)random(1, 10);
    balls[i] = new Ball(x, y, size, c, speed);
}
}

```

```

void draw() {
    background(255);
    for (int i = 0; i < balls.length; i++) {
        balls[i].move();
        balls[i].display();
    }
}

```

```

class Ball {
    int x, y;
    int size;
    color c;
    int speed;

    Ball(int _x, int _y, int _size, color _c, int _speed) {
        x = _x;
        y = _y;
        size = _size;
        c = _c;
        speed = _speed;
    }
}

```

```

void move() {
    x = x + speed;
    if (x > width || x < 0) {
        speed = speed * -1;
    }
}

```

```
void display() {  
    stroke(0);  
    fill(c);  
    ellipse(x, y, size, size);  
}  
}
```